

Chapter - 2 Operators and Expression

Operators are used to perform operations on variables and values.

$$7 + 11 = 18$$

operand ↓ operator ↓ Result

• Types of Operators

- Arithmetic Operators → +, -, *, /, %, ++, --
- Assigned operators → =, +=
- Comparison operator → ==, >=, <=
- Logical operator → &&, ||, !
- Bitwise operator → &, | (operator bitwise)

NOTE: Arithmetic operators cannot work with boolean
%. operators can work with float & double

Simple code for Example

```
Public class OPERATORS
```

```
{
    Public static void main(String[] args)
    {
        int a = 10;
        int b = 10 + a;
        System.out.println("b");
    }
}
```


★ Precedence of Operators ★

The operators are applied and evaluated based on precedence. For example $(+, -)$ has less precedence compared to $(*, /)$. Hence $*$ & $/$ evaluated first.

In case we like to change this order, we use parenthesis.

~~Associativity~~ For Example :

BODMAS code

```
public class BODMAS
{
    public static void main(String[] args)
    {
        int bodmas = 6 * 5 + 10 - 34 / 2;
        System.out.println("bodmas");
    }
}
```

PRECEDENCE CODE

```
Public class PRECEDENCE
{
    public static void main (String[] args) .
    {
        int precedence = 7 * 5 - 45 / 9
        System.out.println (precedence) ;
    }
}
```


M	T	W	T	F	S	S	M	T	W	T	F	S	S	
						1	2	3	4	5	6	7	8	
9	10	11	12	13	14	15	16	17	18	19	20	21	22	
23	24	25	26	27	28	29	30	31						

★ Associativity ★

Associativity tells the direction of execution of operators. It can be left to right or right to left.

* / $\rightarrow$ L to R ;

+ - $\rightarrow$ L to R ;

++, = $\rightarrow$ R to L ;

- Quick Quiz : How will you write the following expression in Java ?

$$\frac{x-y}{2}, \frac{b^2-4ac}{2a}, v^2-u^2, a*b-d$$

4 Programming code

~~public static void~~

Public class QUICKQUIZ

{

public static void main(String[] args)

{

int x = 5;

int y = 3;

int k = (x-y)/2;

System.out.println(k);

}

}

Resulting data type after arithmetic operations

Following table summarizes the resulting data types after arithmetic operations on them.

Result = byte + short $\rightarrow$ Integer

R = short + int $\rightarrow$ Integer

R = long + float $\rightarrow$ float

R = integer + float $\rightarrow$ float

R = character + integer $\rightarrow$ integer

R $\rightarrow$ character + short $\rightarrow$ Integer

R $\rightarrow$ long + double $\rightarrow$ double

R = float + double $\rightarrow$ double

★ Increment and Decrement Operators

a++, ++a $\rightarrow$ Increment operators

$\rightarrow$ First use the value and then increment.

$\rightarrow$ First increment the value then use it.

```
public class Increment
```

```
{
```

```
    public static void main (String [] args)
```

```
{
```

```
        int x = 65;
```

```
        int a = 65 x++;
```

```
        int y = 55;
```

```
        int b = ++y;
```

```
        System.out.println (x++);
```

```
        System.out.println (y);
```

```
        System.out.println (++y);
```

```
        System.out.println (y);
```

```
}
```

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~~NOTE:~~

$a--$, $--a$ → Decrement Operators
 L , $fixot$ $\left\{ \begin{array}{l} \text{use the value and then Decrement} \\ \text{first Decrement the value and then value} \end{array} \right.$

Quick-Quiz : What will be the value of the following Expression (x)

$int\ y = 7;$
 $int\ x = ++y * 8;$
 value of x ?

```
public class Expression
```

```
{
```

```
    public static void main (String[] args)
```

```
    {
```

```
        int y = 7;
```

```
        int x = ++y * 8;
```

```
        System.out.println(x);
```

```
    }
```

```
}
```